Uchiyama's constructive proof of the Fefferman-Stein decomposition

Terence Tao · 23 February 2007

https://terrytao.wordpress.com/2007/02/23/uchiyamas-constructive-proof-of-the-fefferman-stein-decomposition/

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I've just uploaded the short story "Uchiyama's constructive proof of the Fefferman-Stein decomposition". In 1982, Uchiyama gave a new proof of the celebrated Fefferman-Stein theorem that expressed any BMO function as the sum of a bounded function, and Riesz transforms of bounded functions. Unlike the original proof (which relied, among other things, on the Hahn-Banach theorem), Uchiyama's proof was very explicit, constructing the decomposition by building the bounded functions one Littlewood-Paley frequency band at a time while keeping the functions taking values on or near a sphere, and then iterating away the error. Here I have written some notes on how the proof goes. The notes are a little condensed, in that a number of standard computations involving estimations of Schwartz tails, Carleson measures, etc. have been omitted, but hopefully the gist of the argument is still clear.

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[...] Remark 9. The phenomenon of nonlinear quantitative solvability actually comes up in many applications of interest. For instance, consider the Fefferman-Stein decomposition theorem, which asserts that any of bounded mean oscillation can be decomposed as for some , where H is the Hilbert transform. This theorem was first proven by using the duality of the Hardy space and BMO (and by using Exercise 13 from Notes 6), and by using the fact that a function f is in if and only if f and Hf both lie in . From the open mapping theorem we know that we can pick g, h so that the norms of g, h are bounded by a multiple of the BMO norm of f. But it turns out not to be possible to pick g and h in a bounded linear manner in terms of f, although this is a little tricky to prove. (Uchiyama famously gave an explicit construction of g, h in terms of f, but the construction was highly nonlinear; see my blog post on the topic.) [...]



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[...] nature of a certain subspace of a Banach space; see these lecture notes of mine and this old blog post for some discussion.) So one could imagine a sophisticated nonlinear decomposition as a general [...]

 **Matt R.** · 30 October 2015

Dear Professor Tao, the PDF copy of your short story which appears from the link above seems to have some problems in the formatting regarding theorem numbering, equation numbering, etc. Do you have another PDF copy which you can make available? Thank you.

[I don't see the issues with the current PDF, could you be more specific? Thanks – T.]

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[…] theorem). A constructive proof was later given by Uchiyama, and was in fact the topic of the second post on this blog. A notable feature of Uchiyama's argument is that the construction is quite nonlinear; the […]

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[…] with Uchiyama's constructive proof of the Fefferman-Stein decomposition theorem, discussed in this previous post (and in this recent […]